

Software Test Plan (STP)

Web-Based ML Model Evaluator

Project: Web-Based ML Model Evaluator

Version: 1.0

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Based on SRS IDs

ML-EVAL-F, ML-EVAL-NF and ML-EVAL-SR.

1. Introduction

Purpose: This document defines the test plan for the Web-Based ML Model Evaluator. It outlines objectives, scope, strategy, resources, schedule, responsibilities, risks, deliverables, traceability and acceptance conditions.

Scope: Testing covers authentication, CSV dataset upload and validation, dataset preview/profiling, preprocessing, feature/target selection, classification and regression workflows, model selection/configuration, training, testing, evaluation, comparison, visualization, experiment history and result export.

References: Web-Based ML Model Evaluator SRS v1.0; approved design/API specifications; Scikit-learn, Pandas, NumPy, Django and Django REST Framework documentation; database documentation.

Definitions: ML = Machine Learning; SRS = Software Requirements Specification; STP = Software Test Plan; SUT = System Under Test; FR = Functional Requirement; NFR = Non-Functional Requirement; SR = Security Requirement; RTM = Requirements Traceability Matrix; UAT = User Acceptance Testing; API = Application Programming Interface; UI = User Interface.

2. Test Items

- User registration, authentication, logout and session management.
- CSV dataset upload, file validation and dataset ownership.
- Dataset preview, profiling, statistics, missing-value and duplicate detection.
- Data preprocessing, categorical encoding and numerical scaling.
- Feature/target selection and train/test split.
- Classification and regression workflow selection.
- Classification and regression model selection and hyperparameter configuration.
- Model training, test prediction and training-progress feedback.
- Classification and regression evaluation metrics.
- Model comparison, visualization, experiment history and result export.
- REST API, database and file-storage integration.
- Security, reliability, usability, accessibility and performance requirements.

3. Features to be Tested

SRS ID	Feature	Primary Test Case
ML-EVAL-F-001	Registration	TC-AUTH-01
ML-EVAL-F-002	Authentication	TC-AUTH-02
ML-EVAL-F-003	Logout/session invalidation	TC-AUTH-03
ML-EVAL-F-010	CSV upload	TC-DATA-01
ML-EVAL-F-011	Dataset validation	TC-DATA-02
ML-EVAL-F-012	Preview/profiling	TC-DATA-03
ML-EVAL-F-013	Missing values/duplicates	TC-DATA-04
ML-EVAL-F-014	Encoding/scaling	TC-DATA-05
ML-EVAL-F-015	Feature/target/split	TC-DATA-06
ML-EVAL-F-020	Classification/regression workflow	TC-ML-01
ML-EVAL-F-021	Classification models	TC-ML-02
ML-EVAL-F-022	Regression models	TC-ML-03
ML-EVAL-F-023	Hyperparameters	TC-ML-04
ML-EVAL-F-030	Training	TC-TRAIN-01
ML-EVAL-F-031	Test predictions	TC-TRAIN-02
ML-EVAL-F-032	Classification metrics	TC-EVAL-01
ML-EVAL-F-033	Regression metrics	TC-EVAL-02
ML-EVAL-F-034	Controlled training failure	TC-TRAIN-03
ML-EVAL-F-040	Model comparison	TC-COMP-01
ML-EVAL-F-041	Visualization	TC-COMP-02
ML-EVAL-F-042	Experiment history	TC-HIST-01
ML-EVAL-F-043	Result export	TC-EXP-01

ML-EVAL-NF-001	Routine response-time target	TC-PERF-01
ML-EVAL-NF-002	Training feedback	TC-UX-01
ML-EVAL-NF-003	Controlled error handling	TC-REL-01
ML-EVAL-NF-004	Cross-user access isolation	TC-SEC-01
ML-EVAL-NF-005	Responsive/keyboard UI	TC-UX-02

Security requirements:

SRS ID	Security Feature	Test Case
ML-EVAL-SR-001	HTTPS/TLS protection	TC-SEC-02
ML-EVAL-SR-002	Secure password hashing	TC-SEC-03
ML-EVAL-SR-003	Secure upload validation	TC-SEC-04
ML-EVAL-SR-004	Authorization/access control	TC-SEC-05
ML-EVAL-SR-005	No arbitrary code execution	TC-SEC-06
ML-EVAL-SR-006	Sanitized error messages	TC-SEC-07

4. Features Not to be Tested

- Production-scale distributed ML training and autoscaling outside the academic release.
- GPU/deep-learning functionality unless explicitly included in the approved implementation.
- Unrestricted execution of user-supplied Python code.
- AutoML, automated hyperparameter tuning, clustering/unsupervised learning, XAI, model deployment and other future enhancements unless added to scope.
- Internal behavior of third-party libraries beyond the application's integration boundaries.

- Infrastructure behavior outside the supported test environment except where it directly affects a defined requirement.

5. Test Approach / Strategy

The project will use a requirements-driven and risk-based approach. Positive, negative and boundary tests will be used. ML outputs will be checked against independent reference calculations where practical, with fixed random seeds when deterministic results are required.

5.1 Levels

- Unit testing: validation, preprocessing, metric, configuration and service functions.
- Integration testing: frontend/API, API/data pipeline, ML engine, database, storage and export integration.
- System testing: complete browser-to-results workflow using supported datasets.
- Acceptance/UAT: verification of SRS acceptance criteria and intended academic workflow.

5.2 Types

- Functional testing.
- Negative and boundary-value testing.
- Regression testing after defect fixes.
- Performance testing.
- Usability and accessibility testing.
- Security testing.
- ML/data validation testing.

5.3 Entry Criteria

- Stable testable build available.
- Dependencies, database and environment configured.
- Authentication and core dataset/ML workflow available.
- Valid and invalid classification/regression datasets prepared.
- SRS requirement IDs available for traceability.
- Defect/test-result recording mechanism ready.

5.4 Exit Criteria

- 100% of planned acceptance-level tests executed.
- Zero Critical defects open.
- No unresolved High defect blocks a core workflow unless formally approved.
- High-priority requirements verified.
- SR-001 through SR-006 tested with no unacceptable open failures.
- Reference metric checks passed.
- RTM and final Test Summary Report completed.

5.5 Security Validation

- Verify authentication, logout and session invalidation.
- Verify HTTPS/TLS in production-like deployment.
- Verify passwords are securely hashed and not plaintext.
- Reject unsupported, malformed, oversized or unsafe uploads.
- Verify cross-user datasets, experiments and artifacts cannot be accessed.
- Test common injection and unsafe-input conditions.
- Verify user input cannot trigger arbitrary Python execution.
- Verify errors do not expose stack traces, secrets or configuration details.

6. Test Environment

Area	Environment
Hardware	Test laptop/desktop with sufficient CPU, RAM and storage; network access.
OS	Supported Linux or Windows environment.
Backend	Python 3.11+; Django; Django REST Framework.
ML/Data	Scikit-learn; Pandas; NumPy.
Frontend	JavaScript/React or equivalent;

	HTML5/CSS3.
Database	PostgreSQL or MySQL.
Browsers	Chrome, Firefox and Edge.
Tools	pytest/Django TestCase, browser DevTools, Postman/Insomnia, Selenium/Playwright, JMeter/Locust.
Defect Tracking	Jira, GitHub Issues or approved spreadsheet.
Test Data	Valid/invalid CSV datasets for classification and regression, including missing values, duplicates, categorical/numeric fields and boundary cases.

7. Test Schedule Milestone

Milestone	Date	Output
Test plan finalization	17-Sep-2026	Approved STP
Test case design	18-Sep-2026 to 20-Sep-2026	Detailed test cases
Environment setup	21-Sep-2026	Test environment ready
Unit/integration testing	22-Sep-2026 to 26-Sep-2026	Module/interface results
System test execution	27-Sep-2026 to 03-Oct-2026	End-to-end results
Security/performance/usability	04-Oct-2026 to 06-Oct-2026	NFR/security results
Defect fixing/regression	07-Oct-2026 to 10-Oct-2026	Retest/regression results
UAT/final acceptance	11-Oct-2026 to 13-Oct-2026	Acceptance/sign-off

8. Test Deliverables

Deliverable	Contents
Test Plan	Scope, strategy, environment, schedule, roles, risks and acceptance approach.
Test Cases	Functional, negative, ML/data, security and non-functional cases.
Test Scripts	Automated unit/integration/UI scripts where applicable.
Test Data	Valid/invalid CSV files and reference values.
Execution Logs	Results, timestamps, evidence and environment details.
Defect Reports	Defect details, severity, evidence, status and retest result.
RTM	SRS-to-test-case mapping and execution status.
Security/Performance Reports	Measured and observed non-functional results.
Test Summary Report	Coverage, execution, defects, risks and acceptance status.
Final Sign-off	Approved acceptance records.

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	<Name>	Prepare plan, coordinate testing, review coverage, track execution and consolidate results.
Test Engineer	<Name>	Design/execute cases, capture evidence, log defects and perform

		retesting.
Developer / Backend-ML Developer	<Name>	Provide builds, analyze/fix defects and support ML/data validation.
Frontend Developer	<Name>	Fix UI/accessibility/browser issues and support frontend regression.
Project Guide / Product Owner	<Name>	Clarify requirements, review acceptance criteria, provide feedback and approve readiness.

10. Risks and Mitigation

Risk	Mitigation
Stable build delayed	Request early smoke builds and prioritize core workflows.
Environment/database downtime	Maintain reproducible local environment and configuration backups.
Long ML training time	Use controlled datasets and visible training status.
Incorrect evaluation metrics	Use independent reference calculations and fixed random seeds where appropriate.
Unsupported datasets	Document supported-data rules and test invalid inputs.
Unsafe uploads	Validate type/size/content before processing.
Cross-user exposure	Test object-level authorization for datasets, experiments and artifacts.
Dependency/version mismatch	Record exact package versions and reproduce approved environment.

UI regression	Run browser and primary-workflow regression tests after UI changes.
Insufficient evidence	Capture screenshots, logs, API responses and reference calculations for graded tests.

11. Assumptions & Dependencies

- Dependencies and database are available and correctly configured.
- Representative datasets are legally obtained and available for testing.
- Test users can be created for authorization testing.
- Users/testers understand basic ML concepts.
- Initial release focuses primarily on supervised learning and CSV input.
- Server resources are sufficient for approved datasets and models.
- Exact software/library versions used for final testing will be recorded.
- Third-party libraries are tested at the application's documented integration boundaries.

12. Suspension & Resumption Criteria

Suspend testing if:

- The application cannot start or the test build is unavailable.
- Environment/database is unavailable for more than four hours.
- More than 30% of planned tests are blocked by build instability.
- A Critical defect blocks the core dataset-to-evaluation workflow.
- Required test data or critical dependencies are unavailable.

Resume testing if:

- Blocking defects are fixed or an approved workaround exists.
- Environment/database is stable.

- Smoke testing passes on the corrected build.
- Required data, dependencies and configuration are available.
- Affected test cases can be executed again.

13. Test Case Management & Traceability

The RTM maps SRS requirements to test cases and execution status. Status values may be recorded as N = Not started, P = Partially executed/implemented, and A = Accepted/verified.

Requirement	Module/Area	Test Case(s)
ML-EVAL-F-001/F-002/F-003	Authentication	TC-AUTH-01 to TC-AUTH-03
ML-EVAL-F-010 to F-015	Dataset/Data Pipeline	TC-DATA-01 to TC-DATA-06
ML-EVAL-F-020 to F-023	Model Selection/Configuration	TC-ML-01 to TC-ML-04
ML-EVAL-F-030/F-031/F-034	Training/Testing	TC-TRAIN-01 to TC-TRAIN-03
ML-EVAL-F-032/F-033	Evaluation	TC-EVAL-01/02
ML-EVAL-F-040/F-041	Comparison/Visualization	TC-COMP-01/02
ML-EVAL-F-042/F-043	History/Export	TC-HIST-01/TC-EXP-01
ML-EVAL-NF-001	Performance	TC-PERF-01
ML-EVAL-NF-002	Training feedback	TC-UX-01
ML-EVAL-NF-003	Reliability	TC-REL-01
ML-EVAL-NF-004	Access isolation	TC-SEC-01
ML-EVAL-NF-005	Usability/Accessibility	TC-UX-02
ML-EVAL-SR-001 to SR-006	Security	TC-SEC-02 to TC-SEC-07

14. Test Metrics & Reporting

- $\% \text{ Executed} = \text{Executed} / \text{Planned} \times 100.$
- $\% \text{ Passed} = \text{Passed} / \text{Executed} \times 100.$
- $\% \text{ Failed} = \text{Failed} / \text{Executed} \times 100.$
- $\% \text{ Blocked} = \text{Blocked} / \text{Planned} \times 100.$
- Requirement Coverage = Requirements with executed tests / Total testable requirements $\times 100.$
- Defect density, defect aging and severity distribution.
- Regression pass rate = Passed regression tests / Executed regression tests $\times 100.$

Reports: daily/iteration execution status; defect status and aging; security report; performance report; ML/data validation report; RTM/coverage report; Final Test Summary Report; Final Acceptance and Sign-off Report.

15. Approvals

Role	Name	Signature / Date
QA Lead	<Name>	
Developer / Dev Lead	<Name>	
Project Guide / Product Owner	<Name>	
Course Coordinator	<Name>	

Appendix A – Detailed Test Case Register

TC ID	Area	Input/ Condition	Expected Result	Type
TC-AUTH-01	Registration	Valid details	Account created	Functional
TC-AUTH-02	Login	Valid/invalid credentials	Correct accept/reject behavior	Functional/ Negative

TC-AUTH-03	Logout	Authenticated session	Session invalidated	Security
TC-DATA-01	Upload	Valid CSV	Accepted and owned by current user	Functional
TC-DATA-02	Validation	Malformed/ unsupported/ oversized file	Rejected with controlled message	Negative/ Security
TC-DATA-03	Profiling	Valid CSV	Preview/ statistics displayed	Functional
TC-DATA-04	Missing/ duplicates	CSV containing nulls/duplicates	Configured handling applied	Data
TC-DATA-05	Encoding/ scaling	Categorical/ numeric columns	Correct transformations	Data
TC-DATA-06	Feature/ target/split	Valid selections	Correct configuration/ split	Functional
TC-ML-01	Problem type	Classification/ regression	Correct workflow selected	Functional
TC-ML-02	Classification	Supported dataset	Supported models selectable	Functional
TC-ML-03	Regression	Supported dataset	Supported models selectable	Functional
TC-ML-04	Hyperparameters	Valid/invalid values	Accept/reject correctly	Negative
TC-TRAIN-01	Training	Valid configuration	Model trains	Functional

TC-TRAIN-02	Prediction	Trained model/test set	Predictions generated	Functional
TC-TRAIN-03	Training failure	Invalid configuration	Controlled failure	Reliability
TC-EVAL-01	Classification metrics	Known labels/predictions	Matches reference	ML/Data
TC-EVAL-02	Regression metrics	Known targets/predictions	Matches reference	ML/Data
TC-COMP-01	Comparison	2+ experiments	Correct comparison	Functional
TC-COMP-02	Visualization	Completed evaluation	Charts match metrics	Usability
TC-HIST-01	History	Completed experiment	Stored/retrievable by owner	Functional
TC-EXP-01	Export	Completed results	Correct export generated	Functional
TC-PERF-01	Performance	Normal UI/API load	Approved target met	Performance
TC-UX-01	Training feedback	Long-running training	Progress/status visible	Usability
TC-UX-02	Accessibility	Keyboard/browser checks	Primary workflow accessible	Accessibility
TC-REL-01	Reliability	Invalid inputs	No uncontrolled termination	Reliability
TC-SEC-01	Authorization	User A requests User B resource	Access denied	Security

TC-SEC-02	TLS	Protected deployment	HTTPS/TLS verified	Security
TC-SEC-03	Passwords	Inspect storage	No plaintext passwords	Security
TC-SEC-04	Upload security	Unsafe files	Rejected before processing	Security
TC-SEC-05	Object authorization	Cross-user request	Denied	Security
TC-SEC-06	Code execution	Payload-like input	No arbitrary execution	Security
TC-SEC-07	Error disclosure	Forced server error	Sanitized response	Security

Appendix B – Test Execution Record

TC ID	Date	Tester	Result (P/F/B)	Defect ID	Evidence/ Remarks
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Appendix C – Final Test Summary

Planned test cases:

Executed test cases:

Passed:

Failed:

Blocked:

Requirement coverage:

Critical defects open:

High defects open:

Acceptance status: PASS / FAIL / CONDITIONAL

Overall remarks: